First-Order Algorithms for Stochastic Multi-Objective Optimization (SMOO)

Yiyang Li

Department of Applied Mathematics

11 April, 2026

Opening Minds • Shaping the Future
啟迪思維 • 成就未來

Contents

- SMOO Problem
- Solutions of SMOO Problems
- Overview of Current Methods
- A First-order Method for SMOO Problems
- Numerical Experiments
- Conclusions
- References

Contents

- SMOO Problem
- Solutions of SMOO Problems
- Overview of Current Methods
- A First-order Method for SMOO Problems
- Numerical Experiments
- Conclusions
- References

SMOO Problem

We consider the **stochastic multi-objective optimization (SMOO) problem**

$$\min_{\mathbf{x}\in\mathcal{X}} F(\mathbf{x}). \tag{1}$$

The vector-valued function $F : \mathcal{X} \to \mathbb{R}^m$ is defined componentwise by

$$F(\mathbf{x}) = \Big(F_1(\mathbf{x}), \ldots, F_m(\mathbf{x})\Big)^\top, \tag{2}$$

$$F_i(\mathbf{x}) := \mathbb{E}[f_i(\mathbf{x}, \boldsymbol{\xi})], \quad i \in \{1, \ldots, m\}, \tag{3}$$

where

- $\mathcal{X} \subseteq \mathbb{R}^n$ is a nonempty open convex set,
- $\boldsymbol{\xi} : \Omega \to \mathbb{R}^l$ denotes a finite-dimensional random vector defined on a probability space $(\Omega, \mathcal{F}, P)$ with known probability measure P, (e.g.: **random data samples**, noise),
- $f_i : \mathcal{X} \times \Xi \to \mathbb{R}$ are continuously differentiable in $\mathbf{x}$ for any fixed ξ with $L_i(\xi)$-Lipschitz continuous gradients with $\mathbb{E}[L_i(\xi)] < \infty$ and measurable and integrable in ξ for any fixed $\mathbf{x}$.

Contents

- SMOO Problem
- **Solutions of SMOO Problems**
- Overview of Current Methods
- A First-order Method for SMOO Problems
- Numerical Experiments
- Conclusions
- References

Pareto optimality and Pareto stationarity

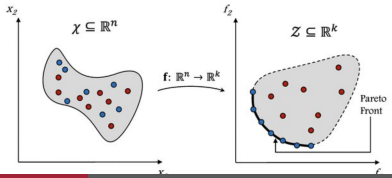
Componentwise order. $\mu \preceq \nu \iff \mu_j \leq \nu_j \; \forall j$, and $\mu \prec \nu \iff \mu \preceq \nu$ and $\mu \neq \nu$.

Definition 1 (Pareto Optimality)

A point $\mathbf{x}^* \in \mathcal{X}$ is a **Pareto optimal** point of $F : \mathcal{X} \to \mathbb{R}^m$, if there does not exist $\mathbf{x} \in \mathcal{X}$ such that

$$F(\mathbf{x}) \preceq F(\mathbf{x}^*) \text{ and } F(\mathbf{x}) \neq F(\mathbf{x}^*).$$

The collection of all such solutions forms the **Pareto set**, while their images form the **Pareto front**.



Definition 2 (Pareto Stationarity)

Let $F : \mathcal{X} \to \mathbb{R}^m$ be continuously differentiable. A point $\mathbf{x}^* \in \mathcal{X}$ is called a **Pareto stationary** point if there exists no direction $d \in \mathbb{R}^n$ such that

$$\nabla F_i(\mathbf{x}^*)^\top d < 0, \quad i = 1, \ldots, m.$$

Equivalently, by Gordan's theorem of the alternative, $\mathbf{x}^*$ is Pareto stationary if there exists $\alpha \in \Delta_m := \{\alpha \geq 0 : \alpha_1 + \ldots + \alpha_m = 1\}$ such that

$$\sum_{i=1}^{m} \alpha_i \nabla F_i(\mathbf{x}^*) = 0.$$

Contents

- SMOO Problem
- Solutions of SMOO Problems
- Overview of Current Methods
- A First-order Method for SMOO Problems
- Numerical Experiments
- Conclusions
- References

SAA + Multi-Gradient Methods

To approximate the SMOO problem (1), we apply the **sample average approximation (SAA) method** (Shapiro, Dentcheva, and Ruszczynski 2021, section 5).

Given i.i.d. samples $\{\xi_j\}_{j=1}^N$, we define the empirical objectives as,

$$F_i^{(N)}(\mathbf{x}) := \frac{1}{N}\sum_{j=1}^{N} f_i(\mathbf{x}, \xi_j), \text{ for } i \in \{1, \ldots, m\}.$$

We then consider the deterministic MOO problem:

$$\min_{\mathbf{x}\in\mathcal{X}} F^N(\mathbf{x}) := \left(F_1^N(\mathbf{x}), \ldots, F_m^N(\mathbf{x})\right)^\top. \tag{4}$$

Then, problem (4) can be solved by algorithms developed for MOO problems.

SAA + Multi-Gradient Methods

At iterate $\mathbf{x}_k$, the **Multi-Gradient Descent Algorithm (MGDA)**(Désidéri 2012) computes a common descent direction at iterate k by solving

$$\mathbf{d}_k = \arg\min_{\mathbf{d}\in\mathbb{R}^n} \left(\max_{i\in\{1,\ldots,m\}} \nabla F_i^N(\mathbf{x}_k)^\top \mathbf{d} + \frac{1}{2}\|\mathbf{d}\|^2 \right). \tag{5}$$

Subproblem (5) is equivalent to

$$\min_{\delta\in\mathbb{R},\, \mathbf{d}\in\mathbb{R}^n} \quad \delta + \frac{1}{2}\|\mathbf{d}\|^2 \quad \text{s.t.} \quad \nabla F_i^N(\mathbf{x}_k)^\top \mathbf{d} \leq \delta, \quad i = 1, \ldots, m. \tag{6}$$

And the dual of (6) is the subproblem as follows,

$$\lambda_k \in \arg\min_{\lambda_k\in\Delta^m} \quad \left\| \sum_{i=1}^m \lambda_k^i \nabla F_i^N(\mathbf{x}_k) \right\|^2, \tag{7}$$

The descent direction at iterate k is $\mathbf{d}_k = -\sum_{i=1}^m \lambda_k^i \nabla F_i^N(\mathbf{x}_k)$.

SAA + Multi-Gradient Methods

The update of $\mathbf{x}$ at the $k+1$-th iterate is $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$.

To determine the step size α_k, a commonly used backtracking method is the **Armijo-type line search** (Wang et al. 2019).

The rule is to choose $\alpha_k > 0$ such that

$$F^N(\mathbf{x}_k + \alpha_k \mathbf{d}_k) \preceq F^N(\mathbf{x}_k) + \gamma \alpha_k J_{F^{(N)}}(\mathbf{x}_k)\mathbf{d}_k, \tag{8}$$

where $\gamma \in (0,1)$, and

$$J_{F^N(\mathbf{x})} = \begin{bmatrix} [\nabla F_1^N(\mathbf{x})]^T \\ \dots \\ [\nabla F_m^N(\mathbf{x})]^T \end{bmatrix} \in \mathbb{R}^{m \times n}.$$

Why SMOO line search is costly?

Key issue. Armijo-type line search needs *reliable function-value comparisons* (in addition to reliable gradients), but these require much larger mini-batches.

- **Small batches** $\Rightarrow$ noisy $F(\cdot)$ $\Rightarrow$ unreliable acceptance / many backtracks.
- **Large batches** $\Rightarrow$ stable $F(\cdot)$ but **high computational cost**.

Quantitative mismatch (Paquette and Scheinberg 2020). Under standard variance bounds, typical "good estimate" requirements imply

$$|S_k^{(g)}| \gtrsim \tilde{\mathcal{O}}\left(\frac{V_g}{\alpha_k^2 \|g_k\|^2}\right), \qquad |S_k^{(f)}| \gtrsim \tilde{\mathcal{O}}\left(\frac{V_f}{\alpha_k^4 \|g_k\|^4}\right),$$

so

$$|S_k^{(f)}| \textit{ is quadratically larger than } |S_k^{(g)}|.$$

Objective-Function-Free Optimization (OFFO)

To address this issue, **objective-function-free optimization (OFFO)** methods have been proposed, notably the stochastic OFFOARp framework (Gratton, Jerad, and Toint 2025). These methods:

- **completely avoid function evaluations,**
- use only stochastic gradients.

In OFFOAR1:

- The step s_k is obtained by minimizing the regularized linear model

$$m_k(\mathbf{s}) := \langle g_k, \mathbf{s} \rangle + \frac{\sigma_k}{2} \|\mathbf{s}\|^2;$$

- The iterate and regularization parameter are updated as

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{s}_k, \qquad \sigma_{k+1} = \sigma_k + \sigma_k \|\mathbf{s}_k\|^2.$$

This framework eliminates line search entirely, motivating its extension to the multi-objective setting.

Contents

- SMOO Problem
- Solutions of SMOO Problems
- Overview of Current Methods
- **A First-order Method for SMOO Problems**
- Numerical Experiments
- Conclusions
- References

Regularized simplex QP and stationarity measures

For a generic matrix $H \in \mathbb{R}^{n\times m}$, define the Tikhonov-regularized simplex QP

$$\lambda_\rho(H) := \arg\min_{\lambda\in\Delta_m} \left\{ \frac{1}{2}\|H\lambda\|^2 + \frac{\rho}{2}\|\lambda\|^2 \right\}, \quad \rho > 0.$$

The associated regularized direction is $d_\rho(H) := -H\lambda_\rho(H)$.

$\rho > 0 \implies$ strong convexity $\implies$ unique multiplier $\lambda_\rho(H)$.

$$\text{Let} \quad G(\mathbf{x}) := \big[\nabla F_1(\mathbf{x}), \ldots, \nabla F_m(\mathbf{x})\big], \quad G^N(\mathbf{x}) = \big[\nabla F_1^N(\mathbf{x}), \ldots, \nabla F_m^N(\mathbf{x})\big].$$

Then the three stationarity measures are:

1. **Unregularized:**
$$\chi(\mathbf{x}) := \min_{\lambda\in\Delta_m} \|G(\mathbf{x})\lambda\|.$$
2. **Regularized:**
$$\chi_\rho(\mathbf{x}) := \|d_\rho(G(\mathbf{x}))\| = \|G(\mathbf{x})\lambda_\rho(G(\mathbf{x}))\|.$$
3. **Regularized SAA target:**
$$\chi_\rho^N(\mathbf{x}) := \chi_\rho\big(G^N(\mathbf{x})\big).$$

Algorithm: SAA-RMGDA

Require: Initial point $\mathbf{x}_0 \in \mathcal{X}$; initial regularization $\sigma_0 > 0$; sample size $N \in \mathbb{N}$; $\rho > 0$.

(SAA data) Draw i.i.d. samples $\{\xi_j\}_{j=1}^N$ once and keep them fixed.

Set $k := 0$ and $\sigma_k := \sigma_0$.

while stopping criterion not met **do**

(SAA gradients) For each $i \in [m]$, set $g_i^N(\mathbf{x}_k) := \frac{1}{N}\sum_{j=1}^N \nabla f_i(\mathbf{x}_k, \xi_j)$, and form $H_k := G^N(\mathbf{x}_k) = [g_1^N(\mathbf{x}_k), \ldots, g_m^N(\mathbf{x}_k)]$.

(MGDA direction) Compute $\lambda_k := \lambda_\rho(H_k)$ and $\mathbf{d}_k := -H_k\lambda_k$.

(AR1 step) Set $\mathbf{s}_k := \frac{1}{\sigma_k}\mathbf{d}_k, \quad \mathbf{x}_{k+1} := \mathbf{x}_k + \mathbf{s}_k$.

(Regularization update) Set $\sigma_{k+1} := \sigma_k + \sigma_k\|\mathbf{s}_k\|^2$.

$k := k + 1$.

end while

Theorem: from SAA to population stationarity

Algorithmic error $\min_{0 \le t \le K} \chi_\rho^N(X_t^N)$

SAA estimation error $\sup_{\mathbf{x} \in \mathcal{K}} \left| \chi_\rho^N(\mathbf{x}) - \chi_\rho(\mathbf{x}) \right|$ → **Regularization bias** $\chi(\mathbf{x}) \le \chi_\rho(\mathbf{x})$

Theorem 1

Fix $\rho > 0$, $N \in \mathbb{N}$, and $K \ge 0$. For the iterates $\{X_t^N\}_{t \ge 0}$ produced by SAA-RMGDA. Let

$$\bar{X}_{N,K} \in \arg \min_{0 \le t \le K} \chi_\rho^N(X_t^N),$$

and assume the iterates satisfy $\{X_t^N : 0 \le t \le K\} \subseteq \mathcal{K} \subset \mathcal{X}$, for some compact set $\mathcal{K}$. then

$$\chi(\bar{X}_{N,K}) \le \chi_\rho(\bar{X}_{N,K}) \le \min_{0 \le t \le K} \chi_\rho^N(X_t^N) + \sup_{x \in \mathcal{K}} \left| \chi_\rho^N(\mathbf{x}) - \chi_\rho(\mathbf{x}) \right|.$$

Quantitative guarantee and parameter scheduling

Theorem 2 (Stationarity complexity for the regularized SAA measure)

Under the smoothness and lower-boundedness assumption on the SAA problem, for every $K \geq 0$,

$$\min_{0 \leq t \leq K} \chi_\rho^N(\mathbf{x}_t) \leq \max\left\{\varepsilon_\rho,\ \sqrt{\frac{\sigma_{\max}(\sigma_{\max} - \sigma_0)}{K+1}}\right\},$$

where

- ε_ρ: *regularization-induced stationarity floor,*
- σ_{max}: *proved upper bound for the adaptive regularization sequence* $\{\sigma_k\}$,
- σ_0: *initial regularization parameter.*

Quantitative guarantee and parameter scheduling

Assumption 1 (finite-sample regularity assumption on $\mathcal{K}$)

For each $i \in [m]$, there exist constants $\bar{M}_{i,\mathcal{K}} < \infty$ and $L_{i,\mathcal{K}} < \infty$ such that, for P-a.e. $\xi \in \Xi$,

$$\sup_{\mathbf{x}\in\mathcal{K}} \|\nabla f_i(\mathbf{x},\xi)\| \le \bar{M}_{i,\mathcal{K}}, \quad \|\nabla f_i(\mathbf{x},\xi) - \nabla f_i(\mathbf{y},\xi)\| \le L_{i,\mathcal{K}}\|\mathbf{x}-\mathbf{y}\| \quad \forall \mathbf{x},\mathbf{y} \in \mathcal{K}.$$

Define $\bar{M}_{\mathcal{K}} := \max_{i\in[m]} \bar{M}_{i,\mathcal{K}}, \quad L_{\mathcal{K}} := \max_{i\in[m]} L_{i,\mathcal{K}}$.

Theorem 3 (Statistical bound for the SAA error on a compact set $\mathcal{K}$)

Fix $\rho > 0$, $\delta \in (0,1)$, and $\varepsilon > 0$. Under Assumption 1, define $\Gamma_G(\mathcal{K}) := \left(\sum_{i=1}^m \bar{M}_{i,\mathcal{K}}^2\right)^{\frac{1}{2}}$, $L_\chi(\mathcal{K},\rho) := 1 + \frac{2\Gamma_G(\mathcal{K})^2}{\rho}$, $\varepsilon_G := \frac{\varepsilon}{L_\chi(\mathcal{K},\rho)}$, $\eta := \frac{\varepsilon_G}{4L_{\mathcal{K}}\sqrt{m}}$. *If*

$$N \ge \frac{32nm\,\bar{M}_{\mathcal{K}}^2}{\varepsilon_G^2}\left(\log\left(\frac{2nm}{\delta}\right) + \log \mathcal{N}(\mathcal{K},\eta)\right),$$

$$\mathbb{P}\left(\sup_{\mathbf{x}\in\mathcal{K}} |\chi_\rho^N(\mathbf{x}) - \chi_\rho(\mathbf{x})| \le \varepsilon\right) \ge 1-\delta.$$

Here $\mathcal{N}(\mathcal{K},\eta)$ denotes the minimal cardinality of an η-net of $\mathcal{K}$.

Quantitative guarantee and parameter scheduling

Scheduling rule

$$\chi(\bar{X}_{N,K}) \lesssim \max\{\varepsilon_\rho, \frac{1}{\sqrt{K}}\} + \text{SAA error}.$$

To target ε-stationarity, choose $K = O(\varepsilon^{-2})$, and N large enough so that the SAA error is $O(\varepsilon)$.

Also, smaller ρ reduces bias, but makes the statistical term harder to control.

Contents

- SMOO Problem
- Solutions of SMOO Problems
- Overview of Current Methods
- A First-order Method for SMOO Problems
- **Numerical Experiments**
- Conclusions
- References

Experiment Settings

Compared Multi-Objective First-Order Methods:

1. SMG (Liu and Vicente 2024): Stochastic with
   - diminishing stepsize $\alpha_k = 0.1 \cdot 0.95^k$,
   - or multi-Armijo line search.
2. Our method:
   - SAA-Regularized Multi-Gradient Descent Algorithm (SAA-RMGDA), which is line-search free.

Setup: All methods initialized at $\mathbf{x}_0 = 0$. All methods use a fixed batch size of N_{train}.

Overview of the task and Dataset 1

Biobjective logistic regression with group-wise regularization

- **Dataset:** Wisconsin Breast Cancer (binary classification).
- Each sample $\xi_j := (a_j, y_j, g_j)$: features a_j, label $y_j \in \{-1, 1\}$, binary group $g_j \in \{1, 2\}$.
- **Model:** Linear classifier $h(\mathbf{x}; a) = z^\top a + b$, $\mathbf{x} = (z^\top, b)^\top$.
- **Group-wise loss:** For group $i \in \{1, 2\}$,

$$f_i(\mathbf{x}, \boldsymbol{\xi}) := \frac{\mathbf{1}_{\{g=i\}}}{p_i} \ell(\mathbf{x}; a, y) + \frac{\lambda_i}{2} \|\mathbf{x}\|^2,$$

  where $\ell(\mathbf{x}; a, y) := \log(1 + \exp(-y(z^\top a + b)))$.
- **Biobjective problem:**

$$\min_{\mathbf{x}} F^N(\mathbf{x}) := \left(F_1^N(\mathbf{x}), F_2^N(\mathbf{x})\right)^\top,$$

  with $F_i^N(\mathbf{x}) = \frac{1}{|J_i|} \sum_{j \in J_i} \ell(\mathbf{x}; a_j, y_j) + \frac{\lambda_i}{2} \|\mathbf{x}\|^2$, $J_i = \{j : g_j = i\}$.
- The objectives are convex but conflicting, reflecting trade-offs between subgroup performances.
- **Evaluation:** 80:20 train-test split. On the test set, we report the classification accuracy.

Numerical Results

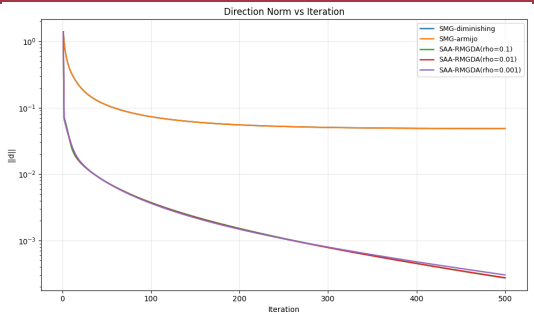


Figure 1: Multi-gradient norm $\|\mathbf{d}_k\|$ versus iteration for SMG with diminishing and multi-Armijo line searches and for the proposed SAA-RMGDA method with different values of ρ on the biobjective logistic regression problem, using a fixed batch size N_{train} for all methods.

method	wall clock time(sec)	train F1	train F2	test accuracy
SAA-RMGDA($\rho = 0.1$)	**0.4945**	0.0833	0.1084	**0.9739**
SAA-RMGDA($\rho = 0.01$)	3.4569	0.0833	0.1084	**0.9739**
SAA-RMGDA($\rho = 0.001$)	4.5698	0.0831	0.1087	**0.9739**
SMG-diminishing	4.3888	0.0986	0.1305	0.9565
SMG-armijo	4.2084	0.0986	0.1305	0.9565

Numerical Results

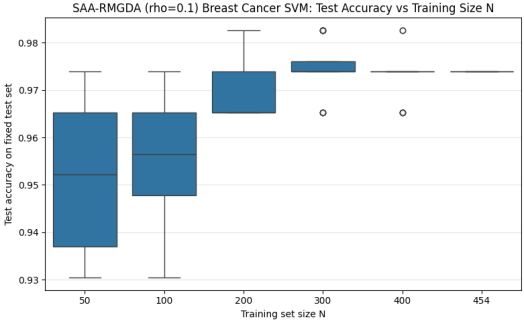


Figure 2: Boxplots of test accuracy for SAA-RMGDA ($\rho = 0.1$) on the breast cancer dataset as the training set size N increases. Accuracy generally improves and becomes more stable with larger training sets.

Overview of the task and Dataset 2

Biobjective portfolio selection with a neural network decision rule

- **Setting:** Stylized mean–variance portfolio selection under i.i.d. market scenarios.
- Each scenario $\xi = (\mathsf{X}, \mathsf{R})$:
  - $\mathsf{X} \in \mathbb{R}^{d \times q}$: asset-feature matrix (features for d assets).
  - $\mathsf{R} \in \mathbb{R}^{d}$: one-period future asset returns.
- **Decision variables $\mathbf{x} = (\vartheta, m)$:**
  - ϑ: neural-network parameters defining the portfolio rule.
  - $m \in \mathbb{R}$: auxiliary reference level in the variance objective.
- **NN portfolio rule:** A Neural Network (with smooth activations) maps features to long-only portfolio weights $w(\vartheta, \mathsf{X}) \in \Delta_d$.
- **Sample-wise objectives for scenario ξ:**

$$f_1(\mathbf{x}, \xi) = -\, w(\vartheta, \mathsf{X})^\top \mathsf{R} \quad \text{(negative return)},$$

$$f_2(\mathbf{x}, \xi) = \big(w(\vartheta, \mathsf{X})^\top \mathsf{R} - m\big)^2 \quad \text{(squared deviation from } m\text{)}.$$

Overview of the task and Dataset 2

- **Biobjective finite-sum problem:**

$$\min_{\mathbf{x}} F^N(\mathbf{x}) = \left(F_1^N(\mathbf{x}), F_2^N(\mathbf{x})\right), \quad F_i^N(\mathbf{x}) = \frac{1}{N}\sum_{j=1}^{N} f_i(\mathbf{x}, \xi_j).$$

- smooth nonconvex problem balancing expected return and risk.
- **Synthetic market simulation:** We generate $N = 100000$ i.i.d. market scenarios. For each scenario and each of the 10 assets, 5 observable features are sampled independently from $[-1, 1]$. Each asset has a latent score computed from its features. A **higher score leads to both higher expected return and higher volatility**. Asset returns are generated from a **factor model** with: a signal-dependent mean return, 3 common market factors, and asset-specific noise.
- **Data & evaluation:** N i.i.d. scenarios, 80:20 train–test split. Report (F_1^N, F_2^N) on the training set and mean/variance of realized portfolio returns on the test set.

Numerical Results

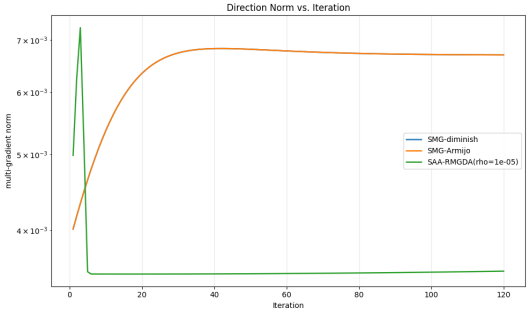


Figure 3: Multi-gradient norm $\|\mathbf{d}_k\|$ versus iteration for SMG with diminishing and multi-Armijo line searches and for the proposed SAA-RMGDA method with $\rho = 1e-5$ on the biobjective financial portfolio management problem, using a fixed batch size N_{train} for all methods.

method	wall clock time(sec)	train F1	train F2	test return	test risk
SAA-RMGDA($\rho = 1e-05$)	**5.3855**	-0.0032	0.0493	0.0017	**0.0495**
SMG-diminish	6.7066	-0.0032	0.0513	0.0017	0.0514
SMG-Armijo	6.1873	-0.0032	0.0513	0.0017	0.0514

Numerical Results

Figure 4: Boxplots of the train-test gaps for the two portfolio objectives across training sizes N. Boxplot of training stationarity measure $\|d_k\|$ versus training set size N for the SAA-RMGDA portfolio problem ($\rho = 10^{-5}$). Larger sample sizes lead to smaller variability and improved training performance.

Contents

- SMOO Problem
- Solutions of SMOO Problems
- Overview of Current Methods
- A First-order Method for SMOO Problems
- Numerical Experiments
- **Conclusions**
- References

Conclusions

- SMOO is challenging because stochastic noise + conflicting objectives make line-search-based methods costly.
- We propose a line-search-free first-order method based on SAA + regularized MGDA + adaptive regularization.
- Theory (via SAA): a modular bridge $\min_k \chi_\rho^N \Rightarrow \min_k \chi_\rho \Rightarrow \min_k \chi$, giving parameter scheduling guidance in (K, ρ, N).
- Experiments show competitive convergence with lower running time.

Contents

- SMOO Problem
- Solutions of SMOO Problems
- Overview of Current Methods
- A First-order Method for SMOO Problems
- Numerical Experiments
- Conclusions
- References

References

- Désidéri, Jean-Antoine (2012). "Multiple-gradient descent algorithm for multiobjective optimization". *Comptes Rendus Mathematique* 350.5, pp. 313–318.
- Gratton, Serge, Sadok Jerad, and Philippe L. Toint (2025). "A Stochastic Objective-Function-Free Adaptive Regularization Method with Optimal Complexity". *Open Journal of Mathematical Optimization* 6, Article no. 5, 24 p. DOI: 10.5802/ojmo.41.
- Liu, S. and L. N. Vicente (2024). "The stochastic multi-gradient algorithm for multi-objective optimization and its application to supervised machine learning". *Annals of Operations Research* 339.3, pp. 1119–1148. DOI: 10.1007/s10479-021-04033-z.
- Paquette, Courtney and Katya Scheinberg (2020). "A Stochastic Line Search Method with Expected Complexity Analysis". *SIAM Journal on Optimization* 30.1, pp. 349–376. DOI: 10.1137/18M1216250.
- Shapiro, Alexander, Darinka Dentcheva, and Andrzej Ruszczynski (2021). *Lectures on Stochastic Programming: Modeling and Theory*. 3rd. PA, US: SIAM.
- Wang, Jinhua et al. (2019). "Extended Newton Methods for Multiobjective Optimization: Majorizing Function Technique and Convergence Analysis". *SIAM Journal on Optimization* 29, pp. 2388–2421. DOI: 10.1137/18M1191737.

Thank you!